DEPARTMENT OF ELECTRICAL ENGINEERING
Indian Institute of Technology, Indore

EE-646 : Image Processing

Presentation

Crypto-Face: Next-Gen End-to-End Encrypted Face Recognition

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Project Reference

Research Paper

Title: CryptoFace:End-to-End Encrypted Face Recognition

Source: CVPR 2024 ?Open Access)

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Michigan State University

Resources

Github

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Paper

The Privacy Challenge

Biometric Vulnerabilities

- **Immutable Data:** Unlike passwords, your face cannot be changed if compromised.
- **Server-Side Risks:** Centralized databases are prime targets for hackers and insider threats.
- **Regulatory Compliance:** GDPR and other laws demand rigorous protection of biometric data.

The Gap

Current secure systems only encrypt the ***feature vector***. The raw image is often processed in cleartext, leaving a vulnerability window.

Facial Recognition

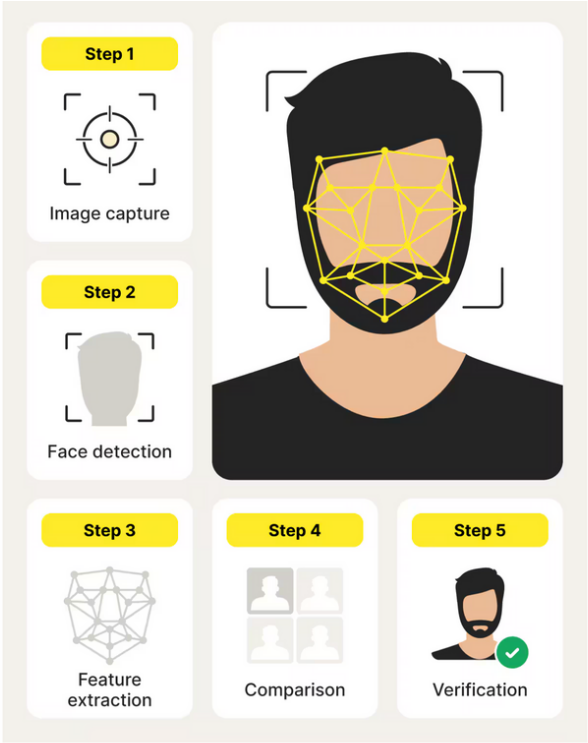


Figure-1

Existing vs. Proposed Solution

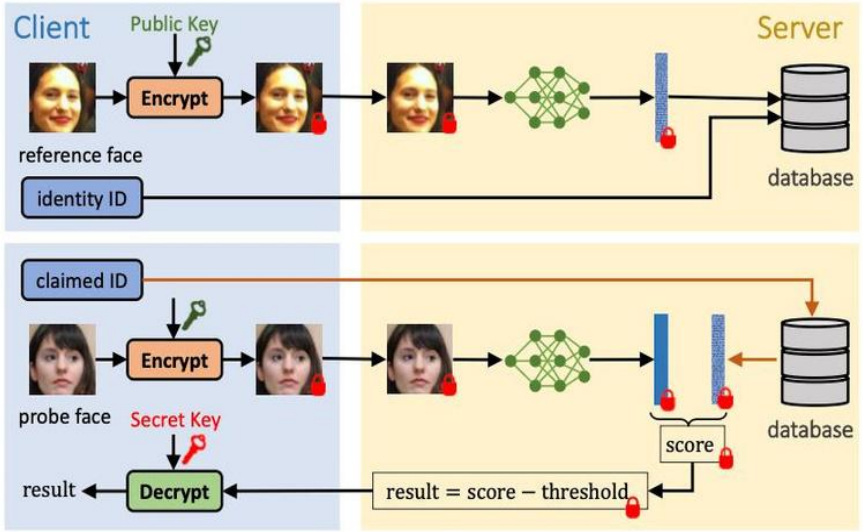


Figure 2. CryptoFace. Top: offline; bottom: online.

Figure-2

Legacy Secure FR

- Client extracts features locally (computationally heavy).
- Only the feature vector is encrypted.
- Vulnerable to "Template Recovery Attacks".

CryptoFace Innovation

- **FullyHomomorphic Encryption (FHE)**

  Compute directly on encrypted data.
- **End-to-End:** Image, Features, and Scores are ***never*** decrypted on the server.
- **Secure Offloading:**

  Client sends encrypted image;Server does the heavy lifting safely.

System Workflow

1. Offline Stage (Enrollment)

Client generates keys and encrypts reference face. Server stores encrypted reference without ever seeing the real face.

2. Online Stage (Verification)

Client encrypts probe face. Server runs the Neural Network on the ciphertext. The result is an encrypted score.

Key Benefit: The server acts as a blind computer. It processes data without knowing

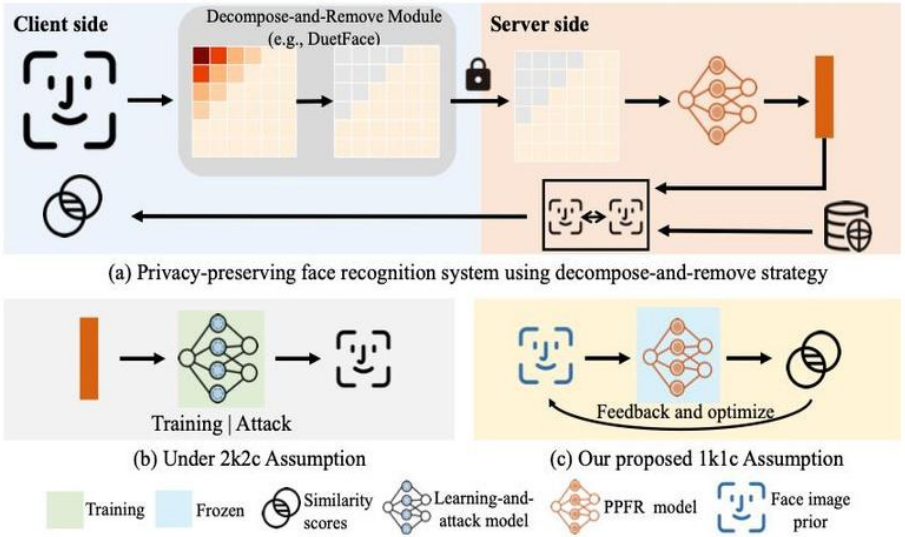


Figure - 3

Crypto-FaceNet Architecture

Patch-Based Design

StandardCNNs are too deep for efficient FHE.

CryptoFaceNet splits the image into non-overlapping patches.

- **Shallow Networks:** Each patch is processed by a shallow "Patch CNN".
- **Lower Latency:** Reduces multiplicative depth (the main bottleneck in FHE?).
- **Global Fusion:** Patch features are aggregated securely to form a global identity vector.

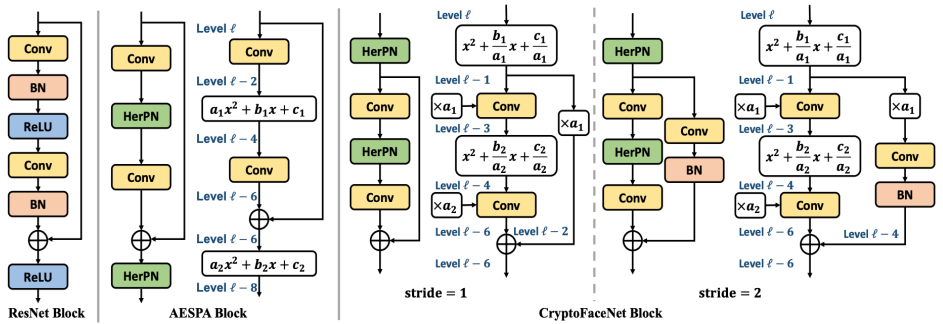


Figure 4. Convolutional blocks and their FHE implementations. Left: ResNet [20]; middle: AESPA [39]; right: CryptoFaceNet.

The Biometric Privacy Challenge

Why Face Recognition Needs Stronger Security

Immutable Data

Unlike passwords, biometric data cannot be changed if compromised — a permanent breach.

Template Recovery

Feature-only encryption is vulnerable — adversaries can reconstruct biometrics from encrypted features.

Centralized Risk

Server databases are prime attack targets. One breach exposes millions of identities.

Inference Attacks

Servers can infer gender, age, race from raw face images processed in cleartext.

GDPR Compliance

EU GDPR and global laws mandate strict biometric data protection.

Scale of Exposure

Billions of face images processed daily across authentication, surveillance & social platforms.

⚡ *Critical Gap: Current secure systems only encrypt the feature vector. The raw face image is processed in cleartext — leaving a dangerous vulnerability window.*

Existing Systems vs. CryptoFace Innovation

❌ Legacy Secure FR

- ✗ Client processes face image locally (cleartext)
- ✗ Only the feature vector is encrypted
- ✗ Raw face image exposed during processing
- ✗ Vulnerable to Template Recovery Attacks
- ✗ Cannot delegate feature extraction to server
- ✗ Limited practical utility — low security



✅ Crypto-Face (CVPR 2025)

- ✓ Face image encrypted BEFORE sending to server
- ✓ Feature extraction on encrypted data (FHE)
- ✓ Matching and scoring fully in encrypted domain
- ✓ Only client can decrypt final result
- ✓ Provable post-quantum security via CKKS
- ✓ 7× faster than state-of-the-art FHE baseline

CryptoFace System Workflow

📋 PHASE 1: OFFLINE (Enrollment)

1 — Generate Keys

Client creates public/secret key pair

2 — Encrypt Face

Reference face encrypted with public key

3 — Send to Server

[encrypted_face] + identity ID

4 — Extract & Store

Server runs FHE inference, stores [feature]

⚡ PHASE 2: ONLINE (Verification) — Latency Critical

1 — Encrypt Probe

Client encrypts probe face + claimed ID

2 — FHE Inference

CryptoFaceNet runs on [encrypted face]

3 — Compute Score

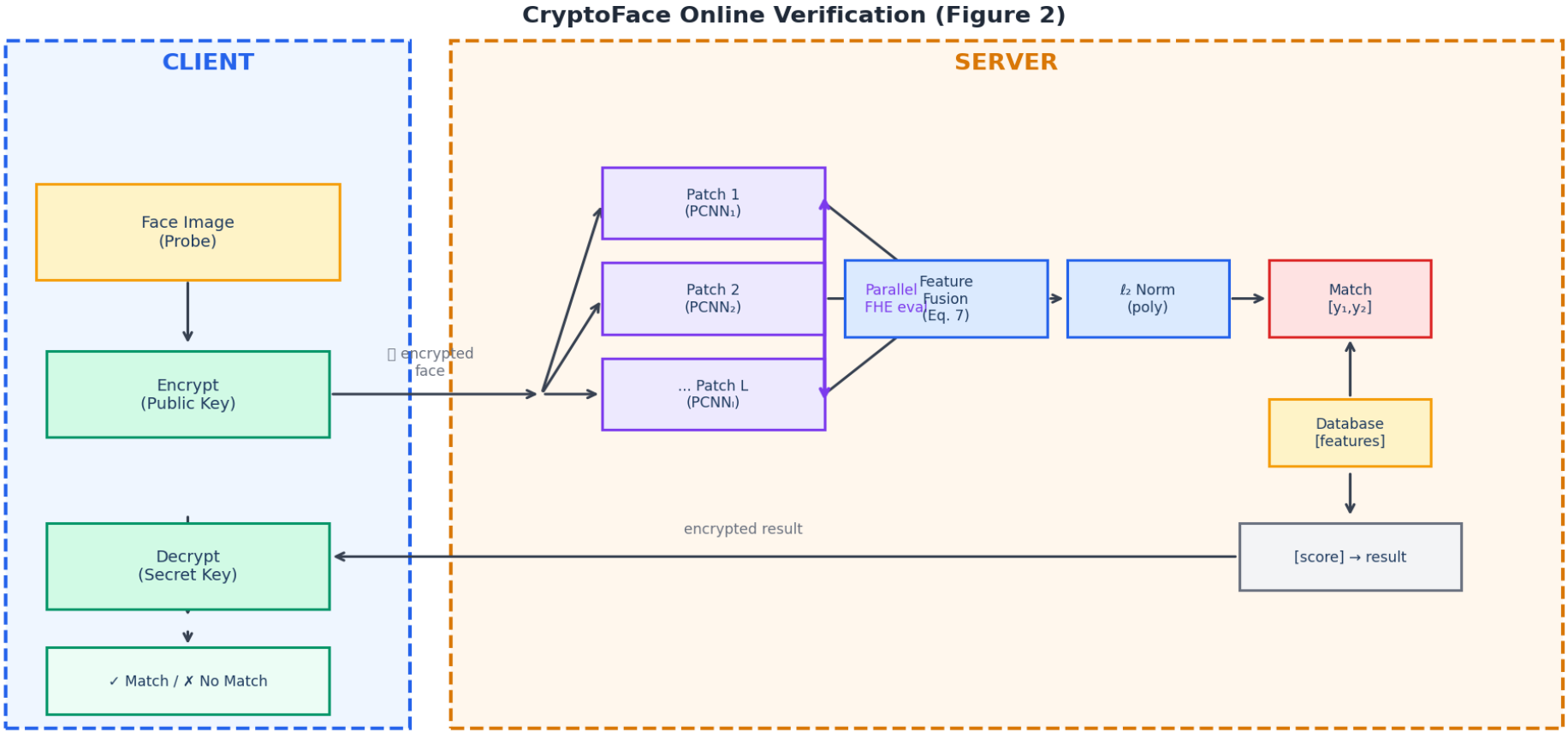
Match([y1],[y2]) in encrypted domain

4 — Decrypt Result

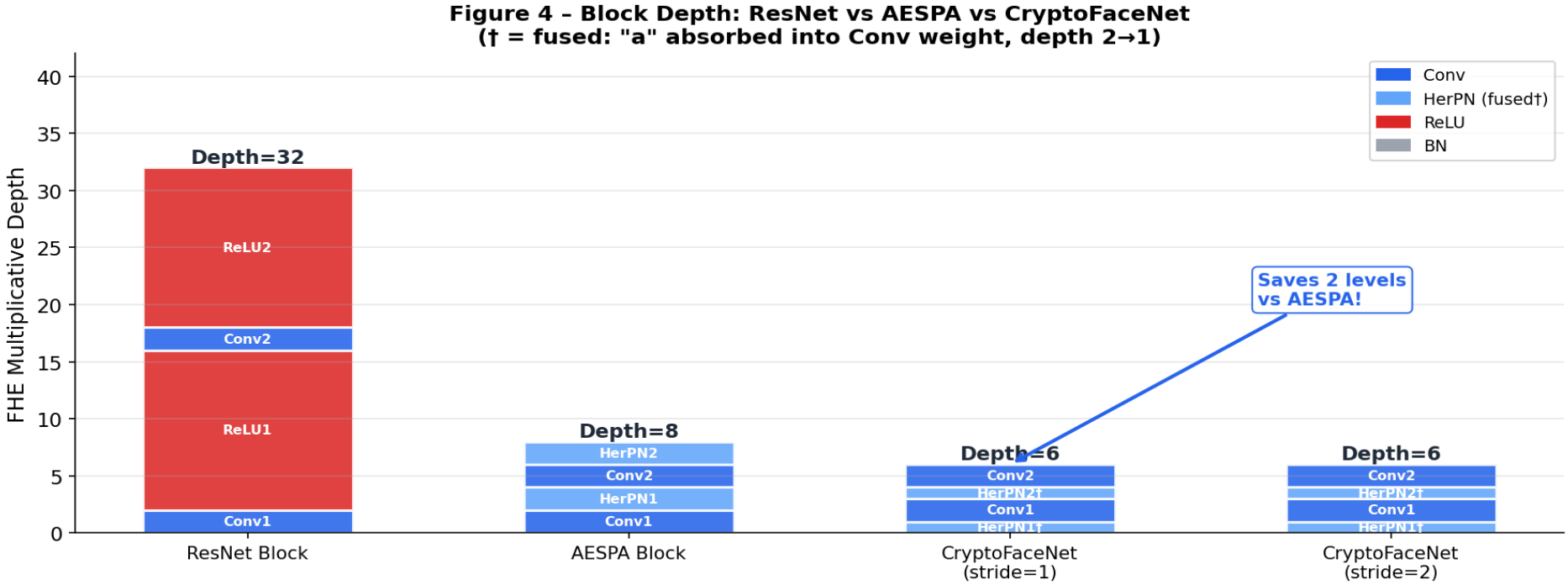
+ve = match, -ve = no match

🔑 ***Key Insight: The server acts as a "blind computer" — it processes encrypted data without ever seeing the original face image, features, or scores. Only the client (holding the secret key) can read the final result.***

Model Architecture



FHE Block Depth Comparison (Figure 4)



ResNet Block

Depth = 32 BN+ReLU — not FHE compatible

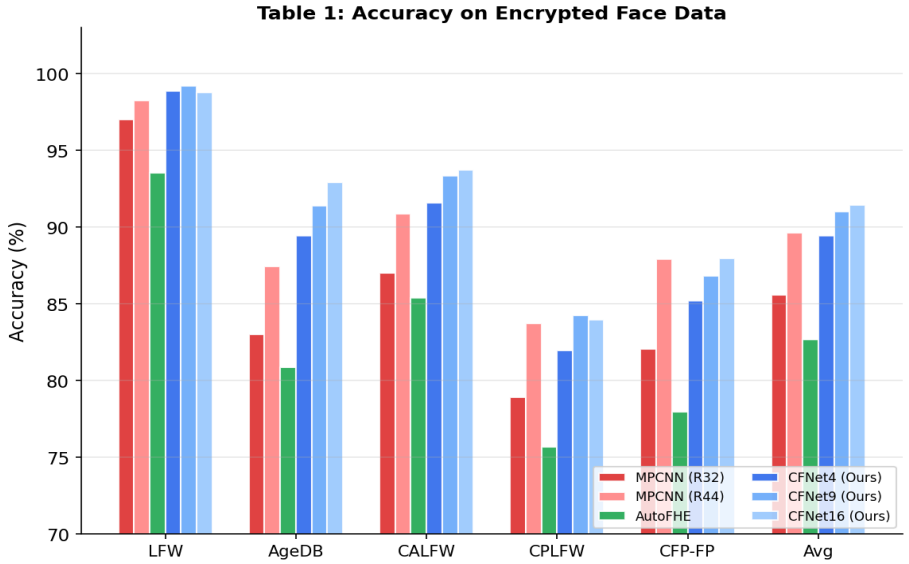
AESPA Block

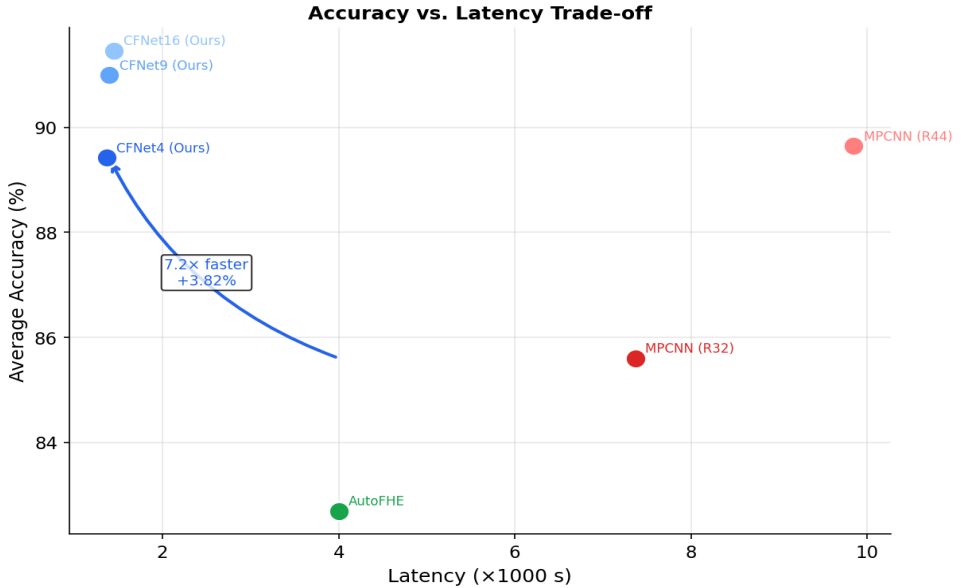
Depth = 8 HerPN full (2 levels each)

CryptoFaceNet Block

Depth = 6 Fuse 'a' → HerPN depth 2→1

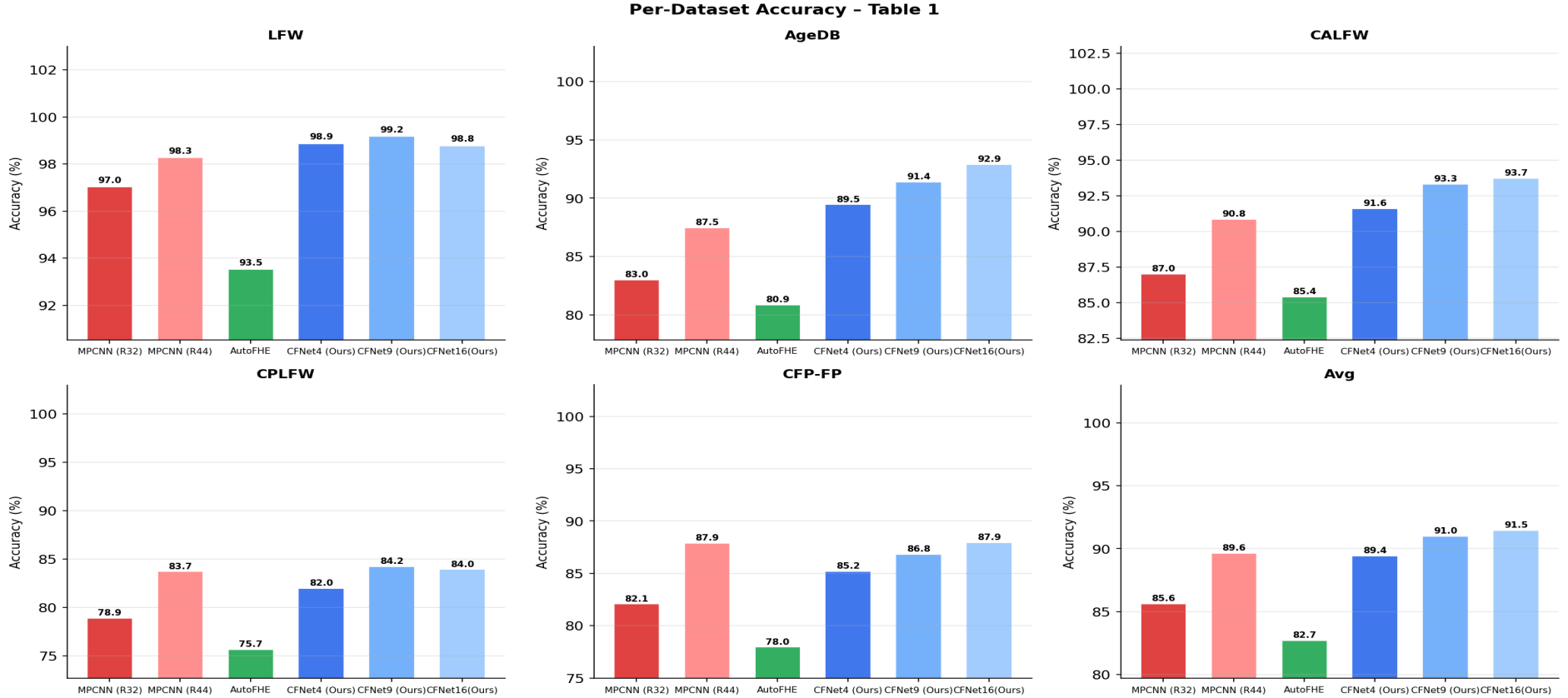
Main Results — Table 1: Accuracy & Latency on Encrypted FHE



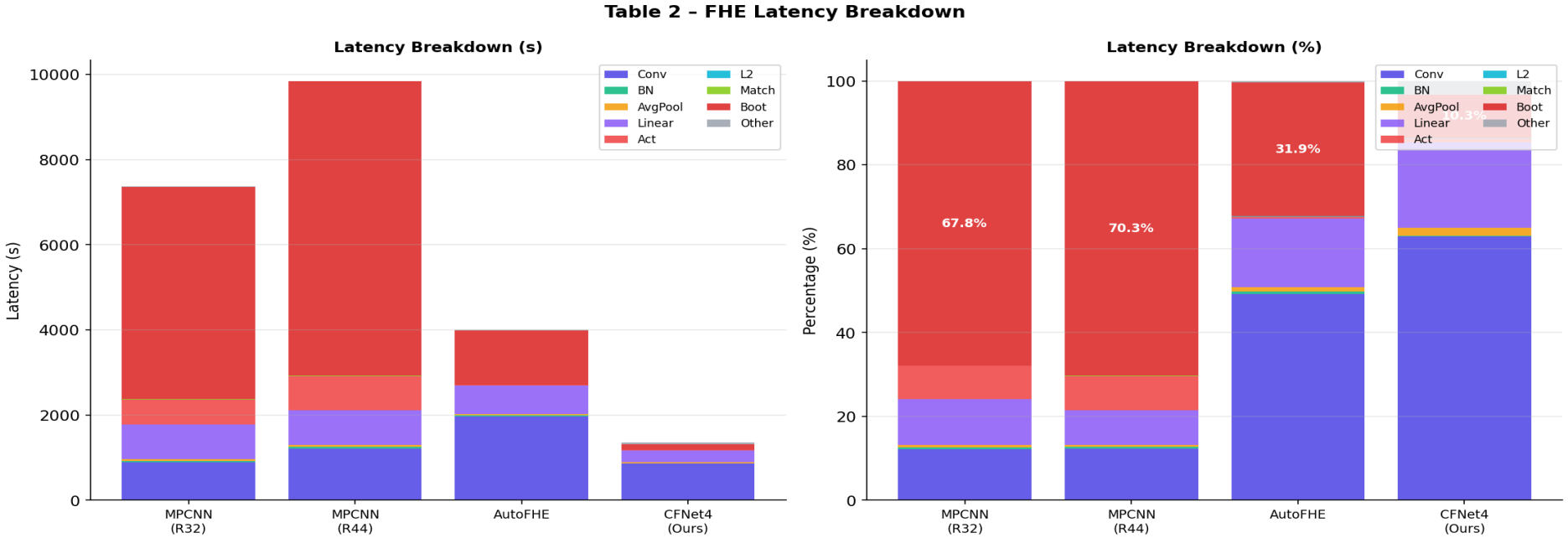


Method	LFW	AgeDB	CALFW	CPLFW	CFP-FP	Avg	Latency(s)	#Boot
MPCNN (R32)	97.02	83.02	87.00	78.90	82.07	85.60	7,367	31
MPCNN (R44)	98.27	87.45	90.85	83.72	87.90	89.64	9,845	43
AutoFHE (R32)	93.53	80.88	85.40	75.67	77.96	82.69	4,001	8
CFNet4 ★	**98.87**	**89.45**	**91.60**	**81.98**	**85.21**	**89.42**	**1,364**	**1**
CFNet9 ★	**99.18**	**91.38**	**93.32**	**84.23**	**86.81**	**90.99**	**1,395**	**1**
CFNet16 ★	**98.78**	**92.90**	**93.73**	**83.95**	**87.94**	**91.46**	**1,446**	**1**

Per-Dataset Accuracy Breakdown — All 5 Benchmarks

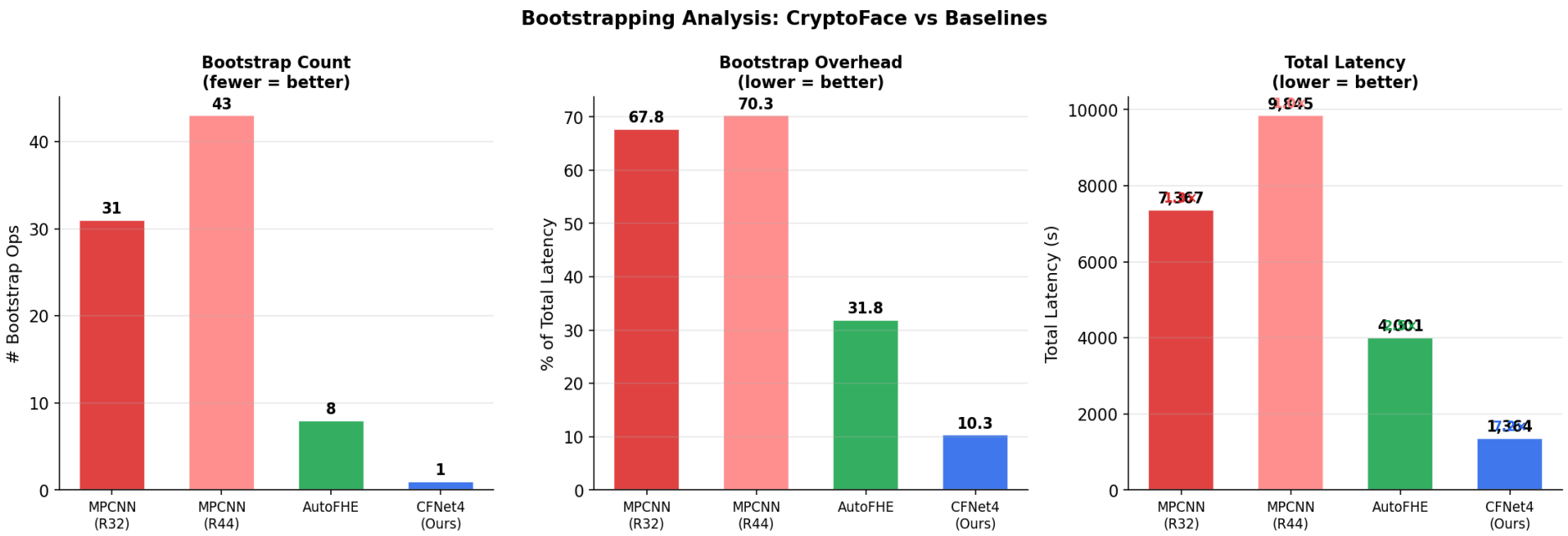


FHE Latency Breakdown — Table 2 (64×64 Encrypted Images)



Method	Conv	Bootstrap	Boot %	Linear	Total (s)
MPCNN (R32)	896s	4,991s	67.76%	807s	7,367
MPCNN (R44)	1,214s	6,917s	70.27%	807s	9,845
AutoFHE (R32)	1,966s	1,274s	31.84%	658s	4,001
CFNet4 ★	**858s**	**141s**	**10.34%**	**277s**	**1,364**

Bootstrapping Analysis — The Key FHE Optimization



43 → 1
Bootstrap Operations
43× reduction vs MPCNN R44

70% → 10%
% of Total Latency
Bottleneck eliminated

7.2×
Speed Improvement
vs MPCNN ResNet44

−17 GB
RAM Savings
269G vs 286G baseline

Implementation — Codebase Structure (Python + PyTorch)

models/cryptofacenet.py

CryptoFaceNet4/9/16 — PatchCNN, FeatureFusion, JigsawHead

models/blocks.py

CryptoFaceNetBlock (depth=6), AESPA (depth=8), ResNet

models/activations.py

HerPN fused (depth=1) / full (depth=2), SquaredActivation

losses/arcface.py

ArcFace additive angular margin loss (s=64, m=0.5)

losses/jigsaw.py

Jigsaw puzzle auxiliary positional loss (α=0.005)

data/dataset.py

WebFace4M training + LFW/AgeDB/CALFW/CPLFW/CFP-FP eval

fhe/fhe_simulator.py

FHE depth tracker, bootstrapping placer, latency estimator

fhe/polynomial_approx.py

Distribution-aware L2 polynomial approx — PolynomialL2Norm

evaluation/evaluator.py

10-fold cross-validation, cosine similarity, AUC/EER

training/trainer.py

SGD + ArcFace + Jigsaw, LR schedule, checkpointing

utils/visualization.py

All 8 paper figures (matplotlib — no GPU required)

scripts/train.py

Main training entry-point — all CLI args, eval, figures

Training Pipeline — Hyperparameters & Commands

Hyperparameter	Value (Paper §4)
Optimizer	`SGD`
Learning Rate	`0.05`
LR Schedule	`×0.1 at epochs [12, 20, 24]`
Epochs	`26`
Batch Size	`256`
Momentum / WD	`0.9 / 0.0005`
Gradient Clip	`1.0`
Patch Size	`32×32`
Feature Dim	`256`

CryptoFaceNet4	
Resolution:	64×64
Patches (L):	4
Parameters:	0.94M
FHE Latency:	1,364s
Avg Accuracy:	89.42%

CryptoFaceNet9	
Resolution:	96×96
Patches (L):	9
Parameters:	2.12M
FHE Latency:	1,395s
Avg Accuracy:	90.99%

Security Guarantees & Threat Model

👤 Adversarial CLIENT

⚠️ **Risk: Tries to infer encrypted features stored on the server's reference database**

🛡️ **Defense: Feature extraction performed entirely server-side. Client receives ONLY the final encrypted match score (+/- scalar). Cannot infer intermediate features.**

🖥️ Adversarial SERVER

⚠️ **Risk: Tries to harvest biometric data from client-provided encrypted images**

🛡️ **Defense: Server holds only ciphertexts. Without the client's secret key, it cannot decrypt or infer any face images, features, or scores.**

Model: Semi-honest two-party protocol (client + server). At most one party is adversarial. Both follow the protocol but may analyse received data. Crypto-Face provides provable post-quantum security via CKKS Fully Homomorphic Encryption.

Key Contributions

01

First End-to-End Encrypted FR

All operations — feature extraction, matching, score thresholding — performed entirely in the encrypted domain. No decryption at any server stage.

02

CryptoFaceNet — Novel FHE Architecture

Mixture of shallow Patch CNNs with FHE depth=6 blocks (saves 2 levels vs AESPA). Parallel evaluation enables near-constant latency across 64→128px.

03

7× Speed Improvement

CryptoFaceNet4: 1,364s vs MPCNN 9,845s. Only 1 bootstrapping operation vs 43. Bootstrap overhead reduced from 70% → 10% of total latency.

04

+8.8% Accuracy Gain

Outperforms AutoFHE by +6.73% and matches MPCNN ResNet44 accuracy at 7× the speed. LFW reaches 99.18% (CryptoFaceNet9, 96×96 resolution).

Secure · Private · Accurate · Practical — CVPR 2025

Conclusion

Crypto-Face represents a paradigm shift in biometric security. By combining **Fully Homomorphic Encryption** with a novel Patch-based Neural Architecture, it delivers the first practical, accurate, and scalable end-to-end encrypted face recognition system.

Secure. Private. Accurate.

References

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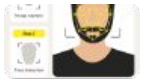
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Image Sources



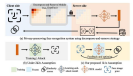
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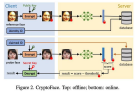
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